1.4.7. Let p be a prime. Prove that order of $GL_2(\mathbb{F}_p)$ is $p^4-p^3-p^2+p$.

proof. Clearly, the Cardinality of $M_{2\times 2}(\mathbb{F}_p)$ is p^4.

And, for any $A \in M_{2\times 2}(\mathbb{F}_p)$,

$$\begin{aligned}\det A = 0 \quad &\text{if and only if} \quad \operatorname{rank} A \leq 1 \\ &\text{if and only if} \quad A = O \ \text{ or } \ \operatorname{rank} A = 1\end{aligned}$$

And, the number of rank $A=1$ matrices is:

$$\underbrace{(p^2-1)}_{\text{number of non-zero vectors}} \cdot \underbrace{p}_{\text{number of multiples}} + \underbrace{p^2-1}_{\text{number of non-zero vectors}} = p^3+p^2-p-1$$

Therefore, $|GL(\mathbb{F}_p)| = p^4-(p^3+p^2-p-1)-1 = p^4-p^3-p^2+p$.

$$p^4-((p^2-1)\cdot p+p^2)$$

$$\times\binom{0}{0} \qquad p-1 \qquad \binom{0}{0} \qquad p^2.$$

⌜1.4.8. $GL_n(F)$ is non-abelian for $n \geq 2$

Proof: Let $A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

$AB = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$, $BA = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. Now, for each $n \geq 2$,

$A_n := \left(\begin{array}{c|c} A & 0 \\ \hline 0 & I \end{array}\right)$, $B_n = \left(\begin{array}{c|c} B & 0 \\ \hline 0 & I \end{array}\right)$

Then $A_n B_n = \left(\begin{array}{c|c} AB & \\ \hline & I \end{array}\right)$, $B_n A_n = \left(\begin{array}{c|c} BA & \\ \hline & I \end{array}\right)$.

Π.4.10. Let $G = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \middle| a, b, c \in \mathbb{R},\ a \neq 0,\ c \neq 0 \right\}$

1). G is closed under the multiplication.

$$\begin{pmatrix} a_1 & b_1 \\ 0 & c_1 \end{pmatrix} \cdot \begin{pmatrix} a_2 & b_2 \\ 0 & c_2 \end{pmatrix} = \begin{pmatrix} a_1 a_2 & a_1 b_2 + b_1 c_2 \\ 0 & c_1 c_2 \end{pmatrix}$$

2). G has inverse

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}^{-1} = \begin{pmatrix} a^{-1} & 0 \\ 0 & c^{-1} \end{pmatrix}$$

3) $G \leq GL_2(\mathbb{R})$

4). $H = \left\{ \begin{pmatrix} a & b \\ 0 & a \end{pmatrix} \middle| a, b \in \mathbb{R},\ a \neq 0 \right\} \leq G$.

Π.4.11.

[1.7. 17. Let G be a group.

G is abelian if and only if $G \to G ; g \mapsto g^{-1}$ is Homomorphism.

Proof This map is clearly bijection. And, $\forall a, b \in G$,

$$ab = ba \iff a^{-1}b^{-1} = b^{-1}a^{-1}$$

Thus proved.

¶1.7.5 Kernel of Action of a group G on A is the same as the kernel of its representation.

proof. Let $G \times A \to A : (g, a) \mapsto g.a$ be an Action. We can consider a Homomorphism:

$$\varphi : G \to S_A : g \mapsto \sigma_g \quad \text{where} \quad \sigma_g : A \to A : a \mapsto g.a$$

By definition, the kernel of given action is a subset of G:

$$\{ g \in G \mid g.a = a, \text{ for all } a \in A \}.$$

And,

$$\ker \varphi = \{ g \in G \mid \varphi(g) = id \} = \{ g \in G \mid \sigma_g = id \} = \{ g \in G \mid g.a = a, \text{ for all } a \in A \}$$

¶1.7.6. Recall that: an Action of G on A is called faithful if:

the induced homomorphism φ is injective.

Prove that group G acts faithfully on A

if and only if

kernel of this action is consisted by only identity.

proof. Trivial.

¶1.7.8. Let A be a non-empty set (not necessarily finite), and let $k \in \mathbb{N}$ s.t. $k < \#(A)$.

Put $\mathcal{B} = \{ \beta \subset A \mid \#(\beta) = k \}$. Define an Action of S_A on $\mathcal{B}$ as:

$$S_A \times \mathcal{B} \to \mathcal{B} : (\sigma, \beta) \mapsto \sigma[\beta]$$

Show that it satisfies Axiom of Action.

Proof. Clearly, $id \in S_A$ satisfies $id[\beta] = \beta$. And, let $\sigma_1, \sigma_2 \in S_A$ be given. Then,

$$(\sigma_1 \circ \sigma_2)[\beta] = \sigma_1[\sigma_2[\beta]] \quad \text{for any } \beta \in \mathcal{B}$$

Therefore it is actually group action.

Example: Consider $A = \{1, 2, 3, 4\}$ and $\mathcal{B} = \{ \beta \subseteq A \mid \#(\beta) = 2 \}$.

Then, $\sigma = (1\ 2)$ acts on $\mathcal{B}$ as:

$$\sigma \cdot \{1,2\} = \{1,2\}, \quad \sigma \cdot \{3,4\} = \{3,4\}, \quad \sigma \cdot \{1,3\} = \{2,3\}, \quad \sigma \cdot \{2,3\} = \{1,3\}$$
$$\sigma \cdot \{1,4\} = \{2,4\}, \quad \sigma \cdot \{2,4\} = \{1,4\}.$$

Group Action.

Def. Let G be a group, and A be a non-empty set.

A map $\alpha : G \times A \to A : (g, a) \mapsto g \cdot a$ is called Group Action if:

1). For all $a \in A$, $1_G \cdot a = a$

2). For all $g_1, g_2 \in G$ and $a \in A$, $(g_1 g_2) \cdot a = g_1 \cdot (g_2 \cdot a)$

Thm. Let $\alpha : G \times A \to A$ be a group action. Then,

$$\varphi : G \to S_A : g \mapsto \sigma_g \quad \text{where } \sigma_g : A \to A : a \mapsto g \cdot a$$

is Homomorphism. Conversely, if $\varphi : G \to S_A$ is Homomorphism, then

$$\alpha : G \times A \to A : (g, a) \mapsto \varphi(g)(a)$$

is group action

「1.7.17. Let G be a group. Consider an Action of G on G as:

$$G \times G \to G ; (g, x) \mapsto gxg^{-1}$$

For fixed $g \in G$, $x \mapsto gxg^{-1}$ is Automorphism on G.

Deduce that $|x| = |gxg^{-1}|$ for all $x \in G$, and $|A| = |gAg^{-1}|$, for any $A \subseteq G$.

Proof. Let $y \in G$. Then, $g(g^{-1}yg)g^{-1} = y$. Thus surjection.

And, suppose $gx_1g^{-1} = gx_2g^{-1} \Rightarrow x_1 = x_2$, by cancelation, therefore injective.

Moreover, Homomorphism is given by $gx_1g^{-1} \cdot gx_2g^{-1} = gx_1x_2g^{-1}$.

To show $|x| = |gxg^{-1}|$, let $|x| = n \in \mathbb{N}$. Since

$$(gxg^{-1})^n = gx^ng^{-1} = gg^{-1} = 1, \quad |gxg^{-1}| \leq n.$$

And, if $(gxg^{-1})^k = 1$ for some $k < n$, then

$$(gxg^{-1})^k = gx^kg^{-1} = 1 \quad \text{implies} \quad x^k = g^{-1}g = 1.$$

which is Contradiction.

To show $|A| = |gAg^{-1}|$, define a map

$$A \to gAg^{-1} ; a \mapsto gag^{-1}.$$

Then, it is bijective (not necessary Homomorphism).

「2.1.6. Let G be an abelian group. Show that the Torsion subgroup

$$T \overset{\text{def}}{=} \{x \in G \mid |x| < \infty\} \subseteq G$$

is subgroup of G,

Proof. Since the identity has order of 1, $T \neq \emptyset$.

Let $x, y \in T$ be given. Since G is abelian, for $n = |x|$ and $m = |y|$,

$$(xy^{-1})^{nm} = x^{nm} \cdot (y^{nm})^{-1} = 1 \cdot 1^{-1} = 1$$

Therefore $xy^{-1} \in G$. By the subgroup Criterion, $T \leq G$.

「2.1.11. Let A, B be groups. Then,

$$\{(a, 1) \mid a \in A\}, \{(1, b) \mid b \in B\} \leq A \times B$$

and

$$\{(a, a) \mid a \in A\} \leq A \times A.$$

Proof. Clear.

21.4
Give an example s.t $H \subseteq G$ is a subset of a group with closed under the operation, but it is not a subgroup.

Sol). $G = (\mathbb{Q} \setminus \{0\}, \times)$ and $H = \{\frac{1}{n} \mid n \in \mathbb{N}\}$

21.6. Let G be a group. The subset

$$T = \{ g \in G \mid |g| < \infty \}$$

is a subgroup when G is abelian. Gives an example T is not subgroup of G.

Sol). $|g_1 g_2^{-1}| = lcm(|g_1|, |g_2|) < \infty \quad \forall g_1, g_2 \in T$ if G is abelian.

Let $G = GL_2(\mathbb{Q})$. Then, $A = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \in GL_2(\mathbb{Q})$ have order 2,

But $AB = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has infinite order. $\left((AB)^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \cdots (AB)^3 = \begin{pmatrix} 1 & 2^2 \\ 0 & 1 \end{pmatrix}, \cdots \right)$

⌈2.2.2. $C_G(Z(G)) = G$. Sol) $C_G(Z(G)) = \{g \in G \mid \forall a \in Z(G),\ gag^{-1} = a\} = G$.

⌈2.2.3. If $A, B \subseteq G$ with $A \subseteq B$, then $C_G(B) \leq C_G(A)$.

Sol). Let $x \in C_G(B)$. Then, $\forall b \in B$, $xbx^{-1} = b$, therefore $\forall a \in A \subseteq B$, $xax^{-1} = a$. Thus $x \in C_G(A)$.

⌈2.2.6. Let $H \leq G$.

a). Prove $H \leq N_G(H)$ and give an example that it is not true if H is not subgroup.

b). prove $H \leq C_G(H)$ if and only if H is abelian.

Proof. Let $h \in H$. Then, $hHh^{-1} \subseteq H$ because it is closed under the operation, and

$H \subseteq hHh^{-1}$ because $\forall k \in H$, $h^{-1}kh \in H$ and $hh^{-1}khh^{-1} \in hHh^{-1}$.

Counterexample: take $G = S_3$ and $H = \{(1\ 2), (1\ 3)\}$. Then

$$(1 2)H(2 1) = \{(1 2), (2 3)\} \neq H$$

And, $H \leq C_G(H) \iff \forall h \in H,\ h \in C_G(H)$

$\iff \forall h \in H,\ gh = hg\ \forall g \in G$

$\iff H$ is abelian.

⌈2.2.8. Let $G = S_n$. Find $|G_i|$ where $i \in \{1, \cdots, n\}$

Sol). Let S_n acts on $\{1, \cdots, n\}$. Then, the stabilizer $G_i = \{\sigma \in S_n \mid \sigma(i) = i\}$

⌈2.2.10. Let $H \leq G$ with $|H| = 2$. Prove $N_G(H) = C_G(H)$ and Deduce that $N_G(H) = G \Rightarrow H \leq Z(G)$.

Proof. $C_G(H) \leq N_G(H)$ is trivial, because $x \in C_G(H) \iff \forall h \in H,\ xhx^{-1} = h \Rightarrow xHx^{-1} = H$.

Conversely, since $|H| = 2$, denote $H = \{1, a\}$. Then,

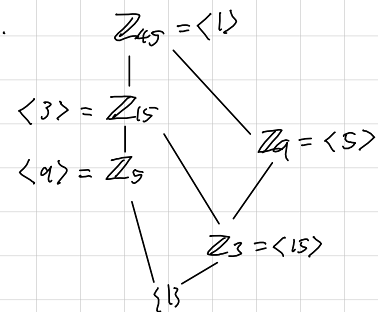
$$x \in N_G(H) \iff xHx^{-1} = H \iff \{1, xax^{-1}\} = \{1, a\} \iff xax^{-1} = a \Rightarrow x \in C_G(H).$$

Thus, if $N_G(H) = G$, then $C_H(H) = G$, thus $H \leq Z(G)$.

⌈2.2.11. $Z(G) \leq N_G(A)$ for any subset $A \subseteq G$.

Proof. Ask a toddler on the street.

2.3.1. Find all Subgroup of $\mathbb{Z}_{45}$.



2.3.2. Let G be a finite group. If $|x| = |G|$, then $G = \langle x \rangle$.
But, it fails when G is infinite.
Proof. Let $|x| = n$. Then, $1, x^1, x^2, \cdots, x^{n-1}$ are distinct elements in G, therefore $G = \langle x \rangle$. Counterexample: $\mathbb{Q}$, $\langle 1 \rangle = \mathbb{Z} \subsetneq \mathbb{Q}$.

2.3.12. Prove $\mathbb{Z} \times \mathbb{Z}$ is <u>Not</u> cyclic.
Proof. Suppose $\mathbb{Z} \times \mathbb{Z}$ is cyclic. Let $(a,b) \in \mathbb{Z} \times \mathbb{Z}$ be a generator.
$n \cdot (a,b) = (1,0)$ for some $n \in \mathbb{N}$ $\Leftrightarrow$ $b = 0$. Thus $(a,b) = (a,0)$
But, it can not generate $(0,1)$.

2.3.16. Let $|x| = n$ and $|y| = m$, and x and y commute.
Then $|xy|$ divides $\mathrm{lcm}(n,m)$.
Proof. Let $k = \mathrm{lcm}(n,m)$. Then $(xy)^k = x^k y^k = 1$. Thus $|xy|$ divides k.

Section 4.1.

#1. Let a group G acts on a set A. $G_a = \{g \in G \mid g \cdot a = a\}$

If $a, b \in A$ and $b = g \cdot a$ for some $g \in G$, then $G_b = g G_a g^{-1}$

Deduce that if G acts transitively on A, then kernel of action is $\bigcap_{g \in G} g G_a g^{-1}$

Proof. Let $x \in G_b$ be given. That is, $x \cdot b = b = g \cdot a$ and $x \cdot b = x \cdot (g \cdot a) = xg \cdot a$.

$g \cdot a = xg \cdot a$ implies $a = g^{-1} x g \cdot a$, therefore $g^{-1} x g \in G_a$.

Conversely, let $ghg^{-1} \in g G_a g^{-1}$ be given. Then,

$ghg^{-1} \cdot b = gh \cdot a = g \cdot a = a$.

Now, if G acts transitively on A, then

$$\text{Kernel of the action} = \bigcap_{b \in A} G_b = \bigcap_{g \in G} g G_a g^{-1}$$

Section 4.1.

#2. Let A be a set, and let $G (\leq S_A)$ acts on A. Prove that

For all $\sigma \in G$ and $a \in A$, $\sigma G_a \sigma^{-1} = G_{\sigma(a)}$.

Deduce that if G acts transitively on A, then $\bigcap_{\sigma \in G} \sigma G_a \sigma^{-1} = 1$.

Proof. Let $\sigma \in G$ and $a \in A$ be given. Then, by the above problem, $\sigma G_a \sigma^{-1} = G_{\sigma(a)}$.

Meanwhile, the kernel of this action is trivial, because

$$\sigma \in \bigcap_{a \in A} G_a \iff \sigma(a) = a \text{ for all } a \in A$$

Now, by the previous problem, $\bigcap_{\sigma \in G} \sigma G_a \sigma^{-1} = 1$.

#3. Let $G \leq S_A$ be a abelian, and transitive. Show that

for all $\sigma \in G \setminus \{1\}$ and $a \in A$, $\sigma(a) \neq a$. Deduce that $|G| = |A|$.

T4.2.8. If $|G:H|=n$, then $\exists K \trianglelefteq G$ s.t. $K \leq H$ and $|G:K| \leq n!$.

Proof. Let π_H be the associated permutation representation afforded by an action

$$G \times \{gH \mid g \in G\} \to \{gH \mid g \in G\} : (x, gH) \mapsto xgH.$$

Then, $\ker \pi_H = \bigcap_{x \in G} xHx^{-1}$, and $\pi_H : G \to S_n$.

Since $\ker \pi_H \trianglelefteq G$ and $\ker \pi_H \leq H$, moreover,

$G/\ker \pi_H \leq S_n$, so $\ker \pi_H$ is the normal we desired.

T4.2.10.

Every non-abelian group of order 6 has a non-normal subgroup of order 2

Proof. Let G be a non abelian group of order of 6.

Suppose that there is normal subgroup N of order of 2

Then, $\ker \pi_N = N$, so $G/\ker \pi_N \cong \mathbb{Z}_3$.